

Aniket Paswan

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🐙 GitHub | 🔗 LinkedIn | 📄 LeetCode

Summary

Computer Science undergraduate with practical experience in backend development, REST API design, API security, and authentication systems. Experienced in designing modular backend solutions, implementing secure authentication workflows, and developing reliable APIs. Strong problem-solving skills with a keen interest in building secure and high-performance software applications.

Education

Bachelor of Technology in Computer Science and Engineering *Expected Graduation: July 2027*

Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh

GPA: 7.79

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Machine Learning, Database Systems, Computer Networks, Software Engineering.

Technical Skills

Languages: Python, Java, SQL, HTML, CSS

Core Fundamentals: Data Structures & Algorithms, Computer Networks, Operating Systems, DBMS

Backend Development: FastAPI, REST APIs, JWT Authentication, SQLAlchemy, API Design

Databases: PostgreSQL, Redis, MongoDB, MS SQL Server

Machine Learning: Pandas, NumPy, XGBoost, Scikit-learn, NLP, Data Preprocessing

Tools: Git, GitHub, Docker, Colab, Locust

Projects

API Security System (Behavior-Based Middleware) | GitHub

May 2026 — Present

FastAPI, Python, REST APIs, Redis, PostgreSQL, SQLAlchemy, React.js

- Developed a behavior-based API security middleware that analyzes incoming requests before business logic execution to mitigate API abuse, credential stuffing, and business workflow attacks.
- Designed a modular API security middleware comprising 10+ independent components including policy evaluation, trust scoring, behavioral analysis, and adaptive enforcement.
- Achieved 16–20 ms average request processing latency using Redis-backed behavioral state management and asynchronous processing.
- Built a React dashboard with 6+ monitoring modules including attack analytics, API usage, suspicious identities, audit logs, policy actions, and API key management.

FastAPI Authentication System | GitHub

Feb 2026 — May 2026

FastAPI, PostgreSQL, Redis, JWT, SQLAlchemy

- Designed modular authentication architecture with role-based authorization, secure password hashing, email verification, and token refresh mechanisms following REST API best practices.
- Reduced database load by approximately 3× using Redis caching, improving API response time under concurrent traffic.
- Successfully handled approximately 800 concurrent users during load testing using Locust.

Mental Health Support Chatbot

Sep 2025 — Dec 2025

PyTorch, Hugging Face, Sentence Transformers, NLP, MongoDB, NumPy

- Developed an AI-powered mental health chatbot using a fine-tuned T5 Transformer model for context-aware response generation.
- Implemented semantic retrieval and safety-aware response generation using Sentence Transformers and sentiment analysis, achieving a 99.2% safety rate.
- Processed conversational datasets into structured context-response pairs through NLP preprocessing and embedding generation for model training.

Certifications

- **Python Programming Certification:** Data structures, and algorithmic problem-solving
- **Generative AI Essentials:** – TCS iON
- **SQL(Basics):** – HackerRank